

Bridging Traditional Health Information Management And Modern Data Intelligence

In A Non-Digitalized Tertiary Hospital

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Abstract

Title:

Bridging Traditional Health Information Management (HIM) and Modern Data Intelligence

Transforming healthcare data in low-resource settings from manual records to intelligent insights.

Many healthcare institutions in low-resource settings, including the University College Hospital (UCH), Ibadan, operate with predominantly manual health information systems. This limits the completeness, quality, and usability of clinical data for operational, research, and decision-support purposes. Effective integration of data engineering, data science, machine learning, and AI can transform the medical data lifecycle, even in non-digitalized environments.

Methods:

We designed and implemented end-to-end data workflows incorporating Django-based ETL pipelines with PostgreSQL and MySQL, Excel preprocessing, and Power BI analytics dashboards. Raw clinical data from paper registers and semi-structured sources were digitized, standardized, and validated. Predictive models using scikit-learn were developed for breast cancer, diabetes risk stratification, and hospital utilization forecasting. Prototype digital health applications were developed using Tkinter and Kivy, and AI-assisted workflows for automated text classification and decision support were implemented. Supplementary YouTube demonstrations document ETL, data science, machine learning, and AI processes.

Results:

The ETL pipelines successfully converted manual hospital data into clean, structured datasets suitable for analysis. Power BI dashboards provided interactive operational insights, including maternal and neonatal outcomes, disease trends, and bed occupancy patterns. Machine learning models achieved predictive accuracy sufficient to support clinical risk stratification and resource planning. Prototype applications demonstrated practical digital interventions that can augment manual HIM workflows.



Conclusions:

Even in non-digitalized hospitals, structured digital workflows integrating data engineering, analytics, and AI can enhance the medical data lifecycle, support evidence-based decision-making, and lay the foundation for scalable digital health transformation. Supplementary demonstration videos provide additional evidence of feasibility and practical implementation.

Keywords: *Medical Informatics Health Information Management (HIM) Medical Data Lifecycle Data Engineering / ETL Machine Learning and AI in Healthcare Health Data Analytics / Visualization*